A Comparative Study of Machine Learning, Deep Learning, and Foundation Models for Short-Term Electricity Load Forecasting

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Abstract

Short-term electricity load forecasting is a core component of smart-grid operation, predictive planning, and industrial energy management. This paper presents a reproducible benchmark comparing SeasonalNaive, a grid-tuned SARIMA baseline, XGBoost, three neural architectures (DLinear, PatchTST, iTransformer), and two time-series foundation models (Chronos-Bolt-Small and TimesFM 2.5). The primary evaluation uses the ECL electricity consumption dataset with aggregate load as the target. ETTh1 is used as a secondary industrial benchmark with HUFL, a load-related transformer variable, as the target; the oil-temperature variable OT is deliberately not treated as electricity load. Experiments cover 24, 96, and 168 hour horizons using a fixed 336 hour context. On ECL, TimesFM obtains the lowest MAE at H=24, while XGBoost is best at H=96 and H=168. Chronos-Bolt is consistently close to the best model without task-specific training. On ETTh1/HUFL, Chronos-Bolt is best at H=24, while DLinear is best at H=96 and H=168. The results show that foundation models are competitive zero-shot forecasters, but classical and supervised models remain strong when training data, feature engineering, and low-latency deployment are available.

Keywords: short-term load forecasting; time series; deep learning; foundation models; SARIMA; XGBoost; smart grid

1. Introduction

Short-term load forecasting (STLF) supports dispatch planning, balancing, tariff-aware operation, maintenance planning, and anomaly-aware industrial energy management. In modern smart-grid and Industry 4.0 settings, forecasting models are expected not only to be accurate, but also to be reproducible, computationally affordable, and robust across operating regimes.

The methodological landscape is broad. Classical methods such as ARIMA-family models and seasonal naive baselines remain important references; machine-learning models such as gradient boosting are attractive because they incorporate lag and calendar features with modest engineering cost; and recent deep-learning and foundation models promise broader transfer across time-series domains. This creates a practical question: when do expensive architectures or zero-shot foundation models actually improve over strong inexpensive baselines?

This paper reports a reproducible execution of the benchmark on two public datasets. The contribution is pragmatic:

- an end-to-end repository with data validation, leakage-aware scaling, metrics, timing, figures, and paper generation;
- real results for SeasonalNaive, grid-tuned SARIMA, XGBoost, DLinear, PatchTST, iTransformer, Chronos-Bolt-Small, and TimesFM 2.5;
- Diebold-Mariano tests computed from per-window errors for the top model pairs;
- an accuracy-latency analysis that separates practical deployment cost from raw accuracy.

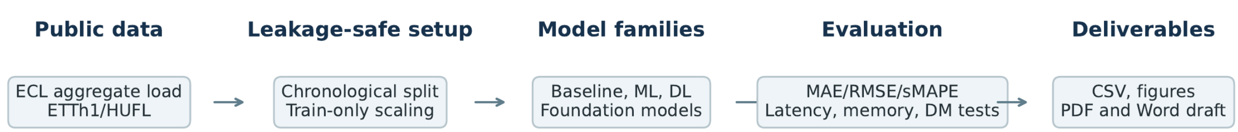

Figure 1: Figure 1. Experimental pipeline used for the benchmark.

2. Related Work

Classical time-series forecasting methods, in particular ARIMA-family models, remain important because they establish interpretable and inexpensive references; automatic order selection by information criteria such as AICc has made them practical to deploy on individual series (Hyndman and Khandakar 2008). Machine-learning approaches based on lagged targets, calendar variables, and gradient-boosted trees are widely used in load forecasting because they are fast to train and robust under tabular feature engineering (Chen and Guestrin 2016).

Deep-learning models such as LSTM (Hochreiter and Schmidhuber 1997), attention-based architectures (Vaswani et al. 2017), Informer (Zhou et al. 2021), DLinear (Zeng et al. 2023), PatchTST (Nie et al. 2023), iTransformer (Liu et al. 2024), and TimesNet (Wu et al. 2023) have broadened the long-horizon forecasting toolbox. Time-series foundation models such as TimesFM (Das et al. 2024) and the Chronos family (Ansari et al. 2024) motivate a new evaluation axis: no-task-specific-training inference versus supervised training on the target dataset.

A fair comparison must state that public benchmarks may have appeared in pretraining corpora. Therefore, zero-shot performance on ECL or ETT should not be interpreted as guaranteed dataset-unseen generalization. The narrower and reproducible claim in this work is that Chronos-Bolt and TimesFM were run without task-specific training in this repository.

3. Methodology

Given a univariate context window, the task is to predict the next H observations for horizons H in {24, 96, 168}. The executed experiments use a fixed context length of L=336 hours, which includes daily and weekly seasonal information and permits use of lag-168 features.

The SeasonalNaive baseline repeats the most recent daily seasonal pattern. The SARIMA baseline uses fixed-grid AICc selection on the training split, then reuses the selected model on test windows via state filtering rather than per-window refitting. XGBoost is trained as a direct multi-output regressor using lag-1, lag-24, lag-168, rolling means over 24 and 168 hours, and cyclic calendar features. DLinear, PatchTST, and iTransformer are trained through NeuralForecast in the same target-only setting, with a short validation-only learning-rate sweep over 1e-3 and 5e-4 plus early stopping. Chronos-Bolt-Small and TimesFM 2.5 are evaluated in zero-shot mode without task-specific training.

All preprocessing for trained models is fitted only on the training split, and metrics are computed after inverse-transforming predictions to the original target scale. Foundation models and SARIMA are evaluated directly on the original target scale.

4. Experimental Setup

The primary dataset is ECL, processed as aggregate hourly electricity load across 321 clients. The secondary dataset is ETTh1 with HUFL as the target. ETTh1 OT is oil temperature and is not used as electricity load. ECL is split chronologically into 70% train, 10% validation, and 20% test. ETTh1 uses the standard 12-month, 4-month, 4-month chronological split.

Accuracy is measured with MAE, RMSE, sMAPE, and wMAPE. Efficiency is measured with training time, single-window inference latency (batch size 1), trainable parameters, and peak GPU memory. XGBoost, DLinear, PatchTST, and iTransformer are evaluated across five seeds (42-46) and reported as mean +/- standard deviation. SeasonalNaive, SARIMA, Chronos-Bolt, and TimesFM are single-run deterministic or zero-shot evaluations and are reported without across-seed standard deviation.

Diebold-Mariano tests are computed on per-window absolute errors for the top two models by MAE at seed 42 for each dataset and horizon. Because flattened point-wise MAE and per-window aggregation weight errors differently, statistical-test rankings can differ slightly from table rankings when models are very close.

5. Results

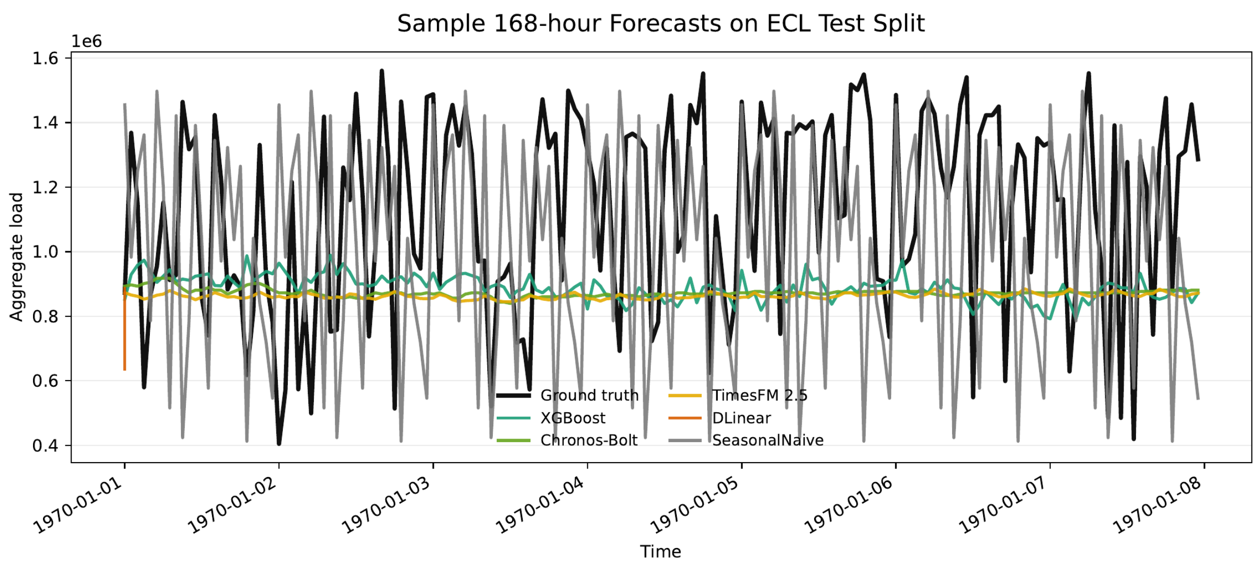

Figure 2: Figure 2. Sample 168 hour ECL forecasts from representative models.

5.1 Accuracy on ECL

On ECL, the strongest group consists of XGBoost and the two foundation models. TimesFM has the lowest point-wise MAE at H=24, but the gap to XGBoost and Chronos-Bolt is below 0.1%, so the practical ranking among the three is essentially flat at the shortest horizon. XGBoost is then marginally best at H=96 and H=168, with Chronos-Bolt close at all three horizons. The grid-tuned SARIMA tracks this top group within a small but consistent gap and remains ahead of the trained neural models.

Model	H=24 MAE	H=96 MAE	H=168 MAE
TimesFM 2.5	**259345.28**	262766.32	265044.60
Chronos-Bolt	259546.11	262601.94	264268.27
XGBoost	259549.38 +/- 13.62	**261792.58 +/- 9.36**	**264220.96 +/- 24.77**
SARIMA	263913.91	267033.48	271294.60
DLinear	278309.93 +/- 78.45	279227.54 +/- 52.48	280272.56 +/- 47.26
PatchTST	279295.35 +/- 274.56	280109.77 +/- 94.31	280876.20 +/- 109.22
iTransformer	283321.32 +/- 278.79	279842.21 +/- 148.37	279476.02 +/- 103.81
SeasonalNaive	338935.90	340658.82	343510.17

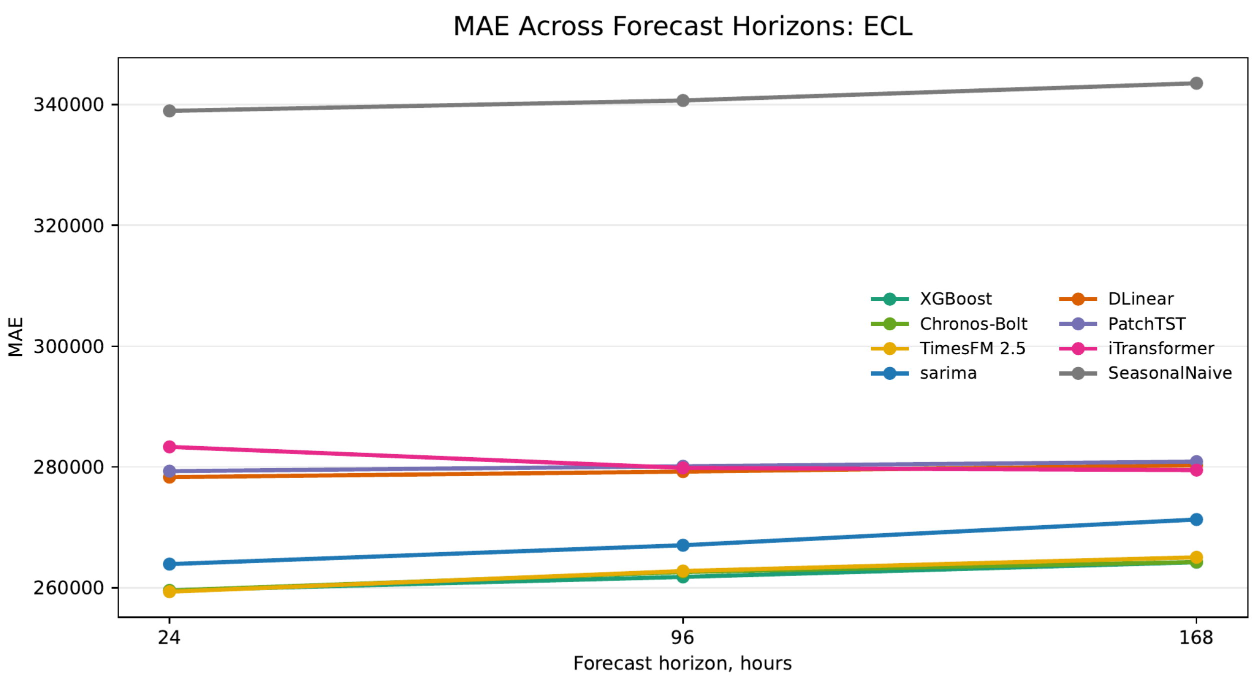

Figure 3: Figure 3. MAE across forecast horizons on ECL.

5.2 Accuracy on ETTh1/HUFL

On ETTh1/HUFL, Chronos-Bolt is best at H=24. PatchTST is the strongest trained model on the same horizon, while DLinear is best at H=96 and H=168. TimesFM is competitive but not best on this benchmark. SARIMA is close to the best classical baseline and beats XGBoost across all three horizons, a role reversal versus ECL where XGBoost dominates SARIMA.

Model	H=24 MAE	H=96 MAE	H=168 MAE
Chronos-Bolt	**3.00**	3.57	3.77
PatchTST	3.04 +/- 0.01	3.62 +/- 0.03	3.95 +/- 0.07
DLinear	3.08 +/- 0.00	**3.55 +/- 0.00**	**3.70 +/- 0.00**
TimesFM 2.5	3.16	3.67	3.79
iTransformer	3.25 +/- 0.02	3.72 +/- 0.01	3.91 +/- 0.02
SeasonalNaive	3.31	3.92	4.18
SARIMA	3.45	3.90	4.18
XGBoost	3.89 +/- 0.01	4.21 +/- 0.01	4.27 +/- 0.01

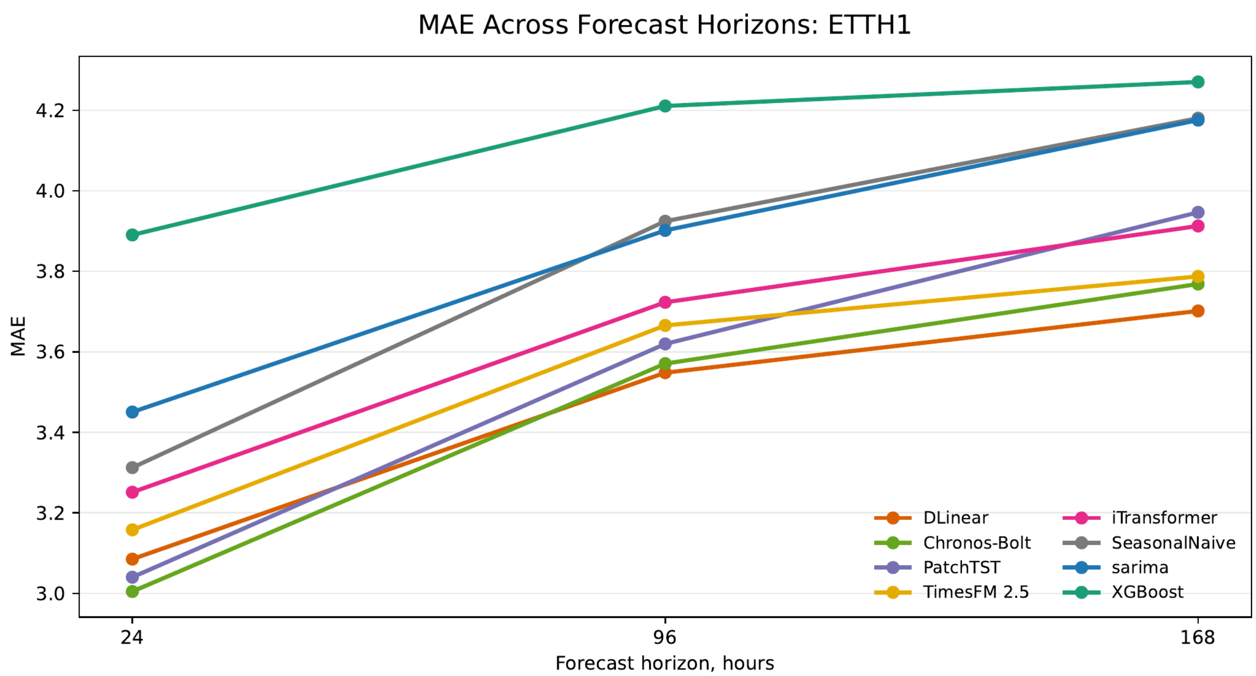

Figure 4: Figure 4. MAE across forecast horizons on ETTh1/HUFL.

5.3 Cross-Dataset View

The normalized heatmap emphasizes that the best model family changes with dataset and horizon. On ECL, XGBoost and the two foundation models form a tight group. On ETTh1/HUFL, Chronos-Bolt and DLinear dominate different horizons. The winner summary provides the same conclusion in a more compact form.

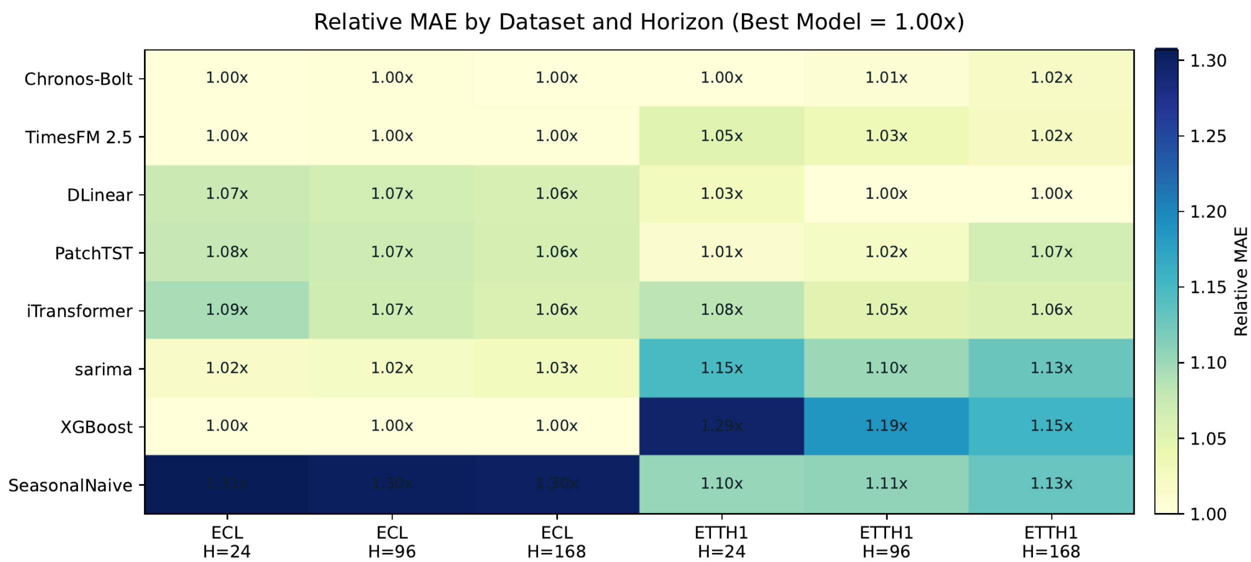

Figure 5: Figure 5. Relative MAE heatmap across datasets and horizons. Values are normalized within each dataset-horizon pair; the best model is 1.00x.

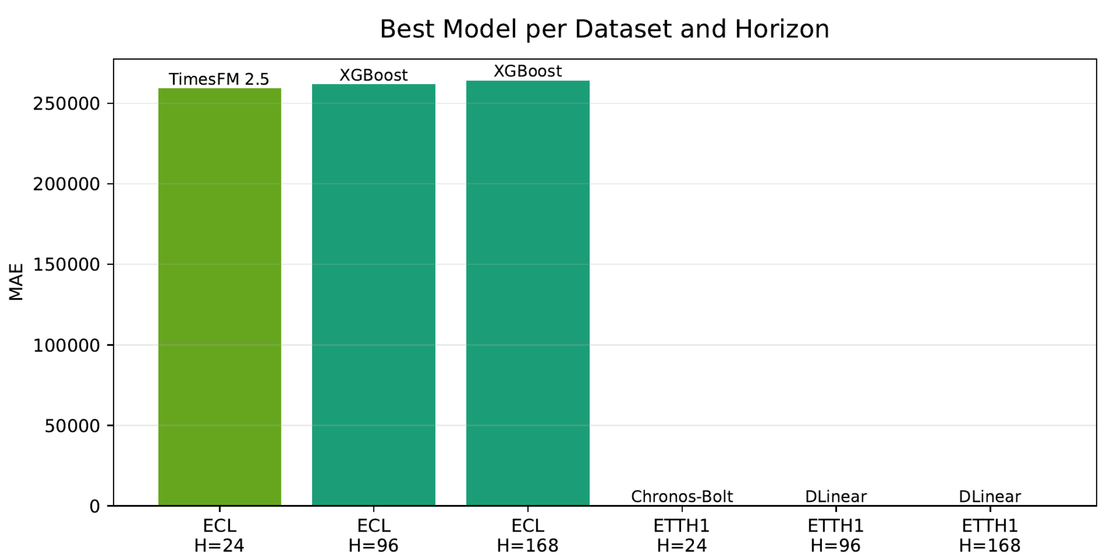

Figure 6: Figure 6. Best model per dataset and horizon.

5.4 Efficiency and Statistical Tests

The efficiency results show the deployment trade-off. SeasonalNaive and SARIMA have the lowest inference latencies. XGBoost remains faster than the neural and TimesFM models and does not require GPU inference. Chronos-Bolt is competitive for zero-shot inference, while TimesFM has the highest latency and GPU memory use in this implementation.

Dataset / H=96	Model	MAE	Inference ms/window	GPU MB
ECL	XGBoost	261792.58	25.09	0.0
ECL	Chronos-Bolt	262601.94	39.64	345.8
ECL	TimesFM 2.5	262766.32	91.99	895.6
ETTh1/HUFL	DLinear	3.55	39.29	29.8
ETTh1/HUFL	Chronos-Bolt	3.57	36.24	345.8
ETTh1/HUFL	XGBoost	4.21	24.47	0.0

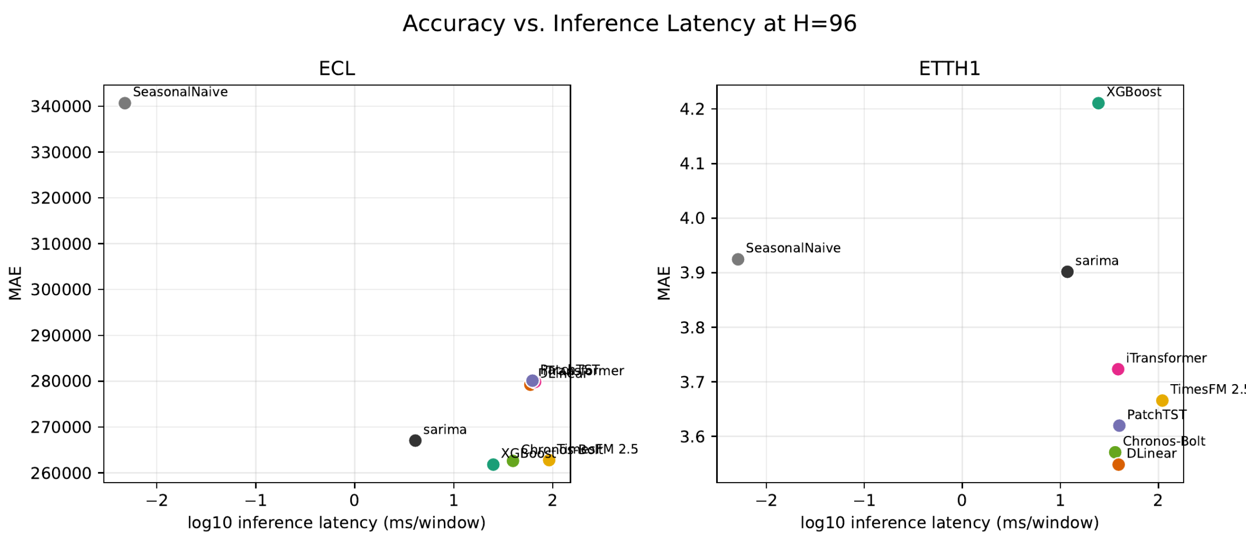

Figure 7: Figure 7. Accuracy-latency view at H=96.

The Diebold-Mariano tests confirm that the top comparisons are not only small numerical differences in flattened MAE. On ECL, the H=96 and H=168 results favor XGBoost over Chronos-Bolt. At H=24, TimesFM and XGBoost are nearly tied in operational terms even though the test rejects equal predictive accuracy. On ETTh1/HUFL, the tests support Chronos-Bolt at H=24 and DLinear at the two longer horizons.

Dataset / Horizon	Comparison	Statistic	p-value	Significant
ECL / H=24	TimesFM vs XGBoost	13.45	$<$0.001	Yes
ECL / H=96	XGBoost vs Chronos-Bolt	-38.15	$<$0.001	Yes
ECL / H=168	XGBoost vs Chronos-Bolt	-2.88	0.004	Yes
ETTh1 / H=24	Chronos-Bolt vs PatchTST	-3.39	$<$0.001	Yes
ETTh1 / H=96	DLinear vs Chronos-Bolt	-2.48	0.013	Yes
ETTh1 / H=168	DLinear vs Chronos-Bolt	-8.81	$<$0.001	Yes

6. Discussion

The results show that model choice is dataset-dependent. On aggregate ECL load, engineered lag features with XGBoost remain an extremely strong and inexpensive option. The grid-tuned SARIMA baseline is also competitive and remains ahead of the trained neural architectures evaluated here. Foundation models are attractive when avoiding task-specific training is valuable: Chronos-Bolt is close to XGBoost on ECL and best at the shortest ETTh1/HUFL horizon. DLinear is the strongest trained neural model in this benchmark and dominates the longer ETTh1/HUFL horizons.

From a deployment perspective, the models occupy different operating regimes. XGBoost is the most attractive choice when a conventional supervised training pipeline is acceptable: it is accurate on ECL, has low single-window latency, and does not require GPU inference. SARIMA is a competitive low-cost backup with no learned features and millisecond inference. Chronos-Bolt is useful when task-specific training is undesirable, for example in cold-start industrial monitoring, rapid prototyping, or sites where historical data cannot be pooled for model training. TimesFM shows strong point accuracy at the shortest ECL horizon, but its latency is substantially higher in this implementation.

The transformer results should be read carefully. PatchTST and iTransformer are credible architectures, but this benchmark uses a compact target-only configuration and a fixed training budget. Their underperformance here does not imply that transformers are generally unsuitable for load forecasting; rather, it shows that they do not automatically beat simpler methods under constrained, reproducible settings.

7. Limitations

The main limitations are public-benchmark pretraining-contamination risk for foundation models, limited hyperparameter search, target-only modeling, and the absence of private industrial deployment data. A rolling AutoARIMA baseline was attempted, but full per-origin grid search was prohibitively expensive in this protocol. The partial run is retained in repository logs but excluded from the main comparison tables. The grid-tuned SARIMA reported here is the practical replacement: orders are selected on the training split by AICc and the parameters are reused for forecast windows, giving a fast and well-defined ARIMA-family baseline.

Chronos-Bolt also emits a warning that prediction lengths above 64 are outside its recommended range. Therefore, H=96 and H=168 Chronos results should be interpreted as practical stress tests rather than optimal use of that model. The foundation-model results should also not be overinterpreted as guaranteed dataset-unseen zero-shot generalization. ECL and ETT are public benchmarks and may overlap with pretraining or evaluation corpora used during model development.

8. Conclusion

We built and executed a reproducible STLF benchmark pipeline on ECL and ETTh1/HUFL. On ECL, XGBoost, Chronos-Bolt, and TimesFM are operationally near-tied across horizons, with TimesFM best at H=24 and XGBoost best at H=96 and H=168. On ETTh1/HUFL, Chronos-Bolt wins H=24, PatchTST is the best trained model at H=24, and DLinear wins H=96 and H=168. These findings support a pragmatic recommendation: compare against strong simple baselines first, use DLinear and PatchTST as neural references, and treat foundation models as valuable cold-start tools rather than universally dominant replacements for supervised forecasting.

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